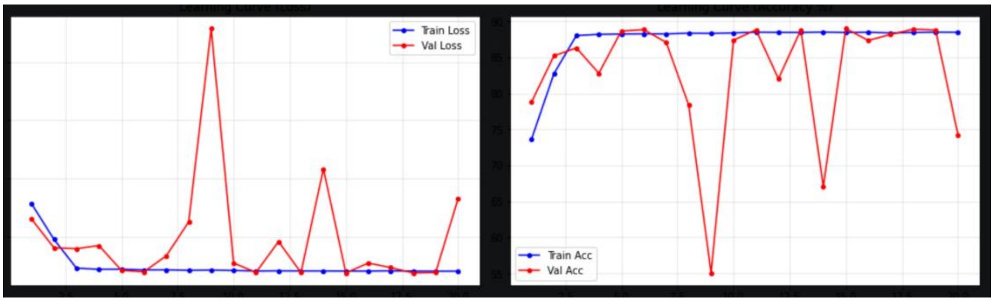
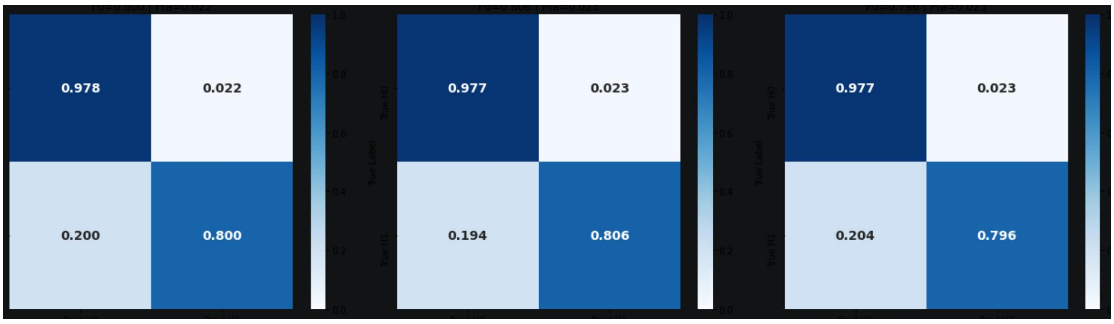
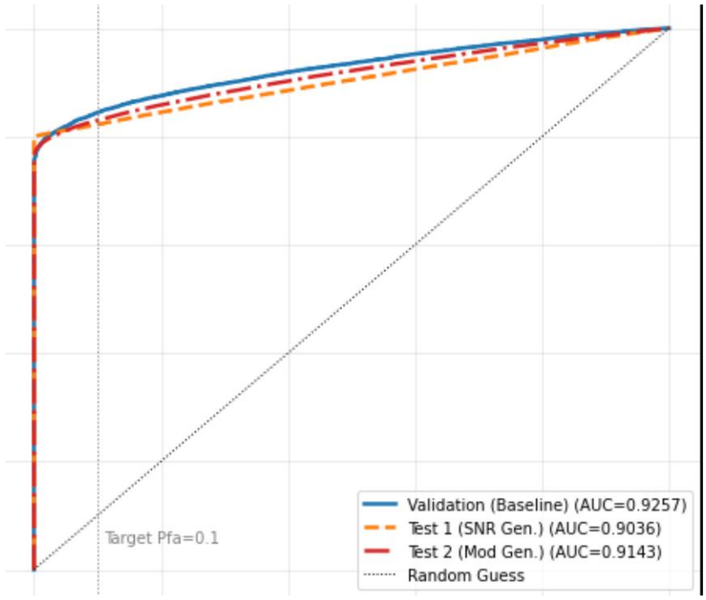
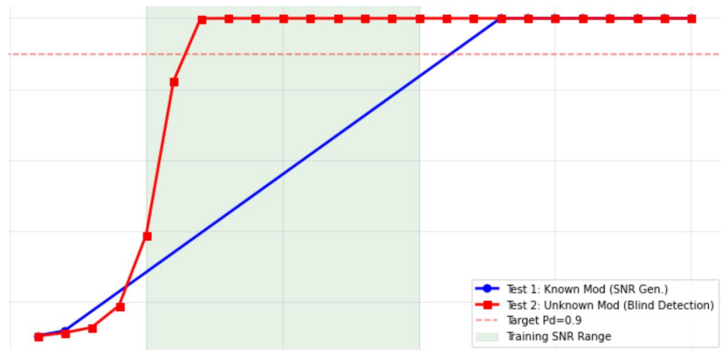
1、BiLSTM

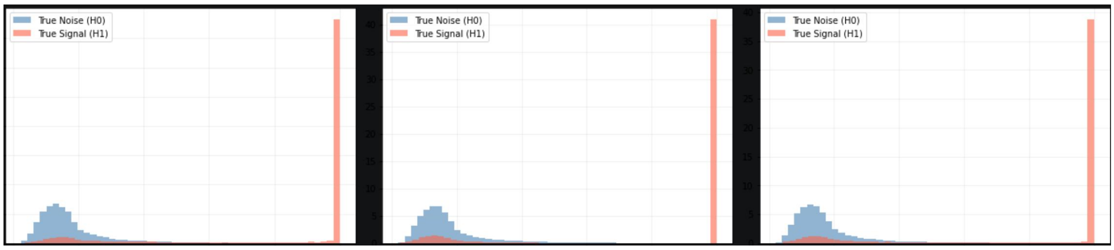
	验证集	Test1	Test2
F1	0.8783	0.8814	0.8751
AUC	0.9257	0.9036	0.9143
Pd	0.8003	0.8059	0.7962
Pfa	0.0221	0.0227	0.0234
ACC	0.8891	0.8916	0.8864
Pre	0.9732	0.9726	0.9714
Rec	0.8003	0.8059	0.7962

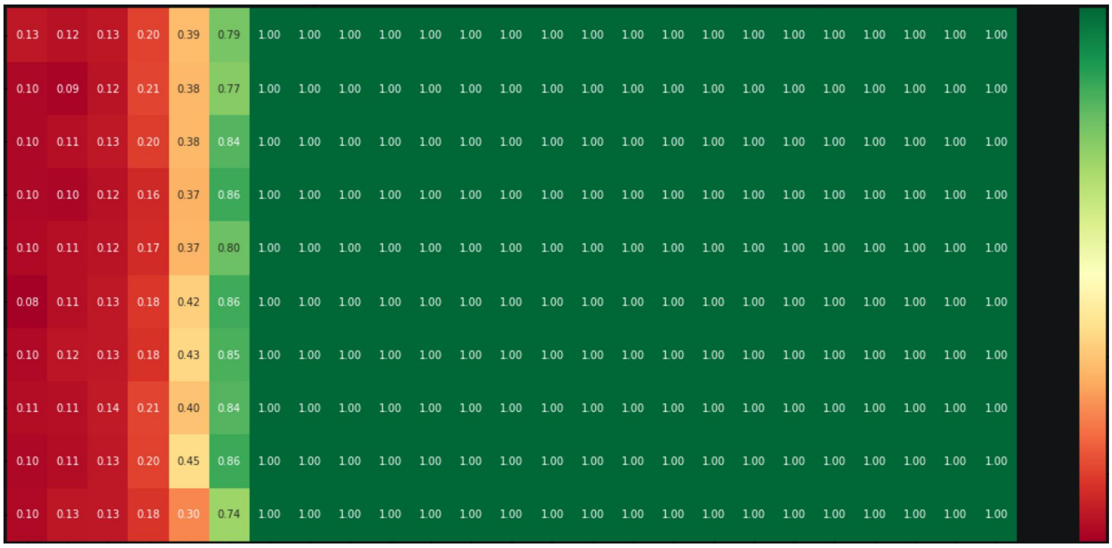






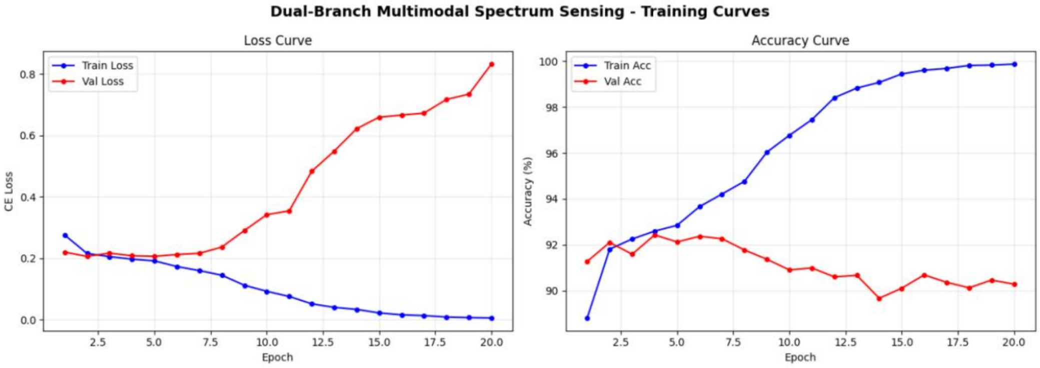




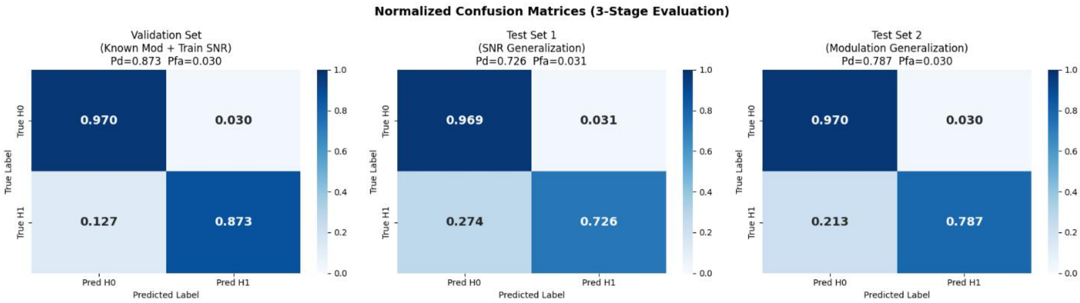


2、8_Image

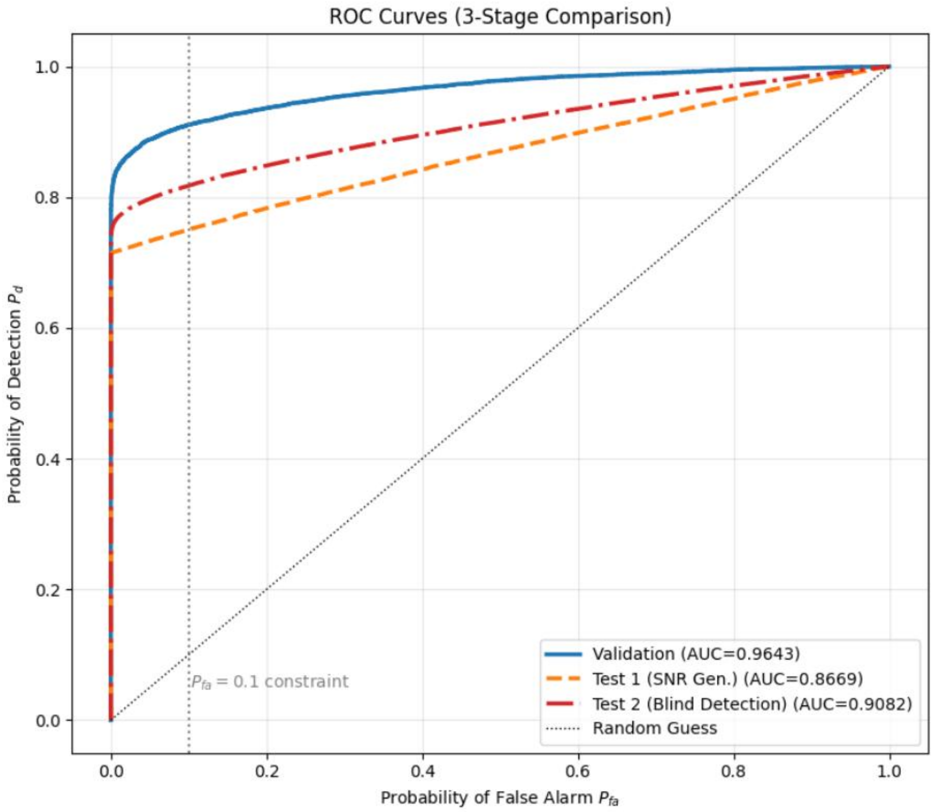
	验证集	Test1	Test2
F1	0.9170	0.8263	0.8658
AUC	0.9643	0.8669	0.9082
Pd	0.8725	0.7260	0.7866
Pfa	0.0304	0.0312	0.0304
ACC	0.9210	0.8474	0.8781
Pre	0.9663	0.9588	0.9628
Rec	0.8725	0.7260	0.7866



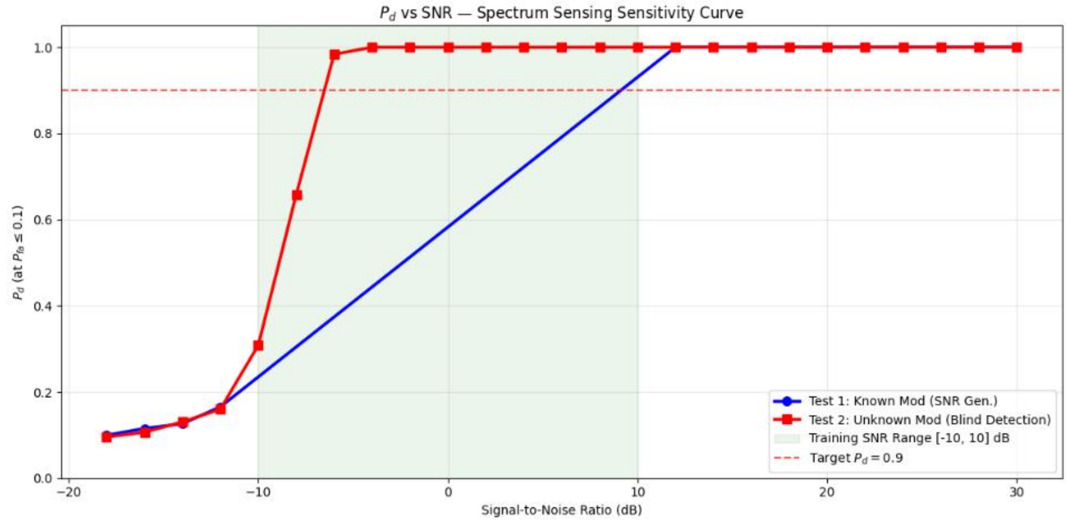
Dual-Branch Multimodal Spectrum Sensing - Training Curves
Loss Curve
Train Loss
Val Loss
CE Loss
Epoch
Accuracy Curve
Train Acc
Val Acc
Accuracy (%)
Epoch

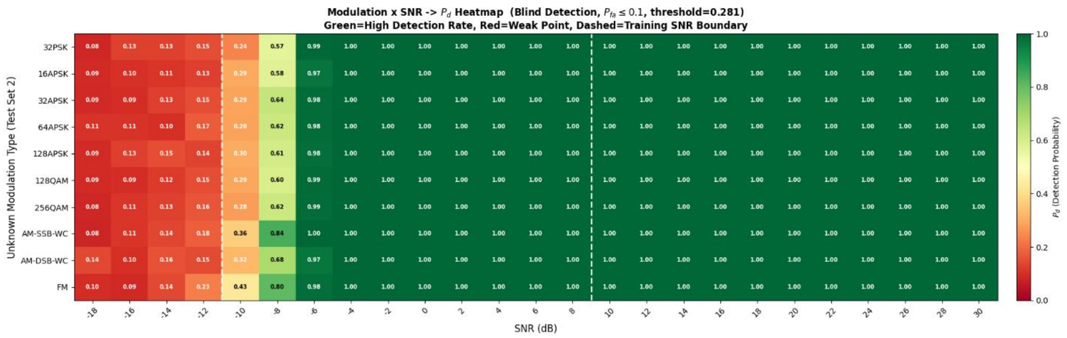


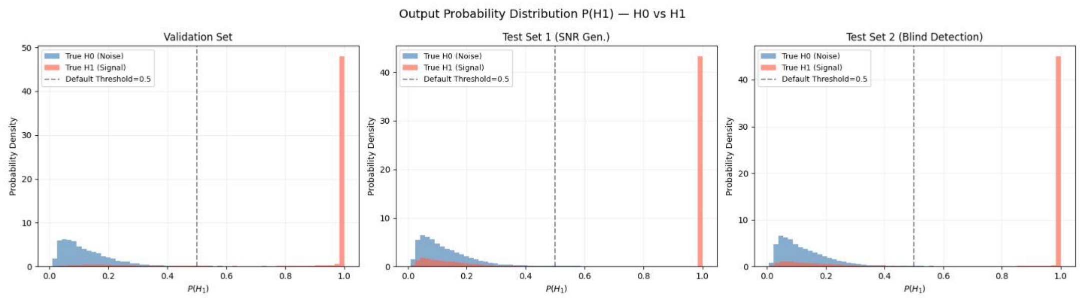
Normalized Confusion Matrices (3-Stage Evaluation)
Validation Set
(Known Mod + Train SNR)
Pd=0.873 Pfa=0.030
0.970
0.030
0.127
0.873
Test Set 1
(SNR Generalization)
Pd=0.726 Pfa=0.031
0.969
0.031
0.274
0.726
Test Set 2
(Modulation Generalization)
Pd=0.787 Pfa=0.030
0.970
0.030
0.213
0.787
True H0
True H1
True Label
Pred H0
Pred H1
Predicted Label



ROC Curves (3-Stage Comparison)
Probability of Detection P_d
Probability of False Alarm P_{fa}
$P_{fa} = 0.1$ constraint
Validation (AUC=0.9643)
Test 1 (SNR Gen.) (AUC=0.8669)
Test 2 (Blind Detection) (AUC=0.9082)
Random Guess

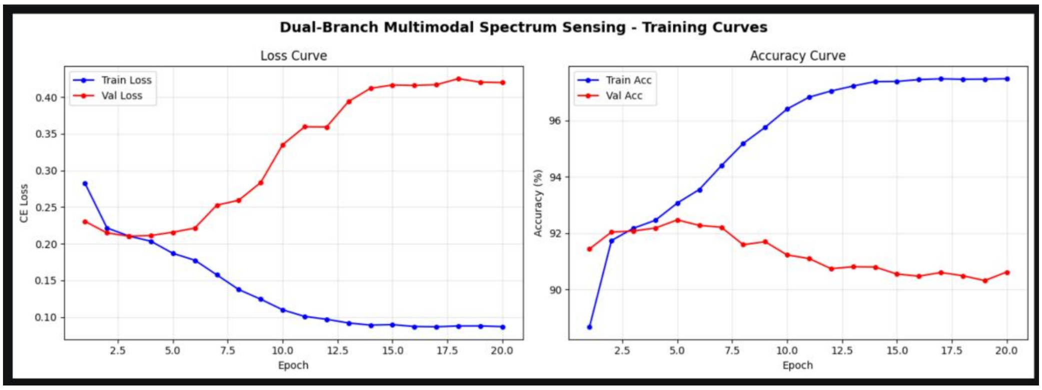


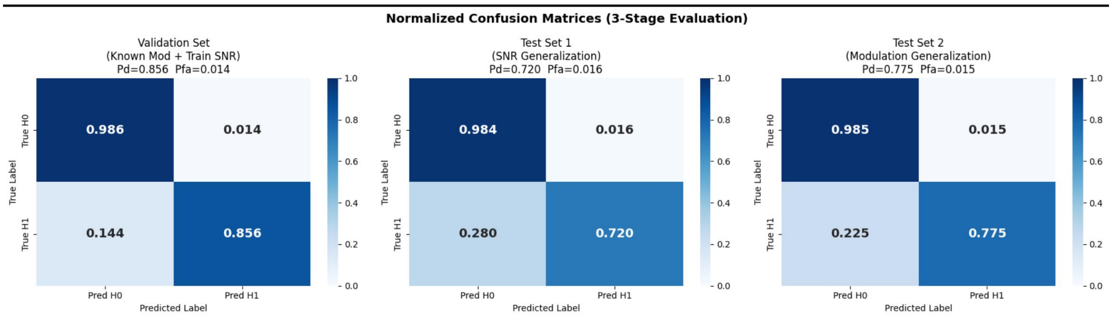


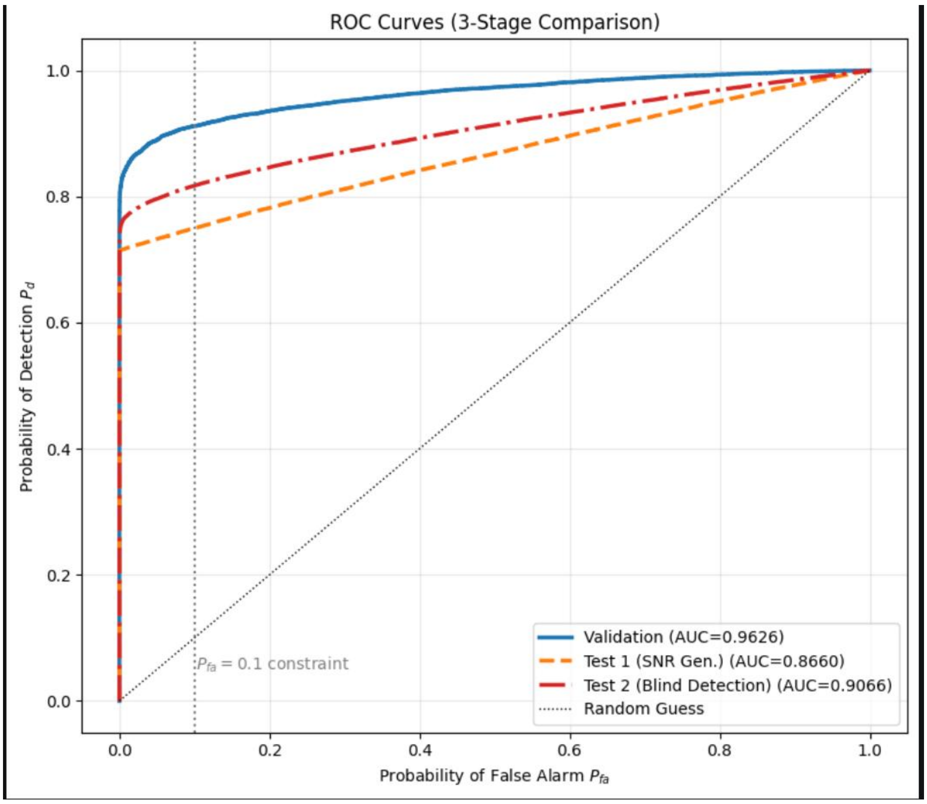


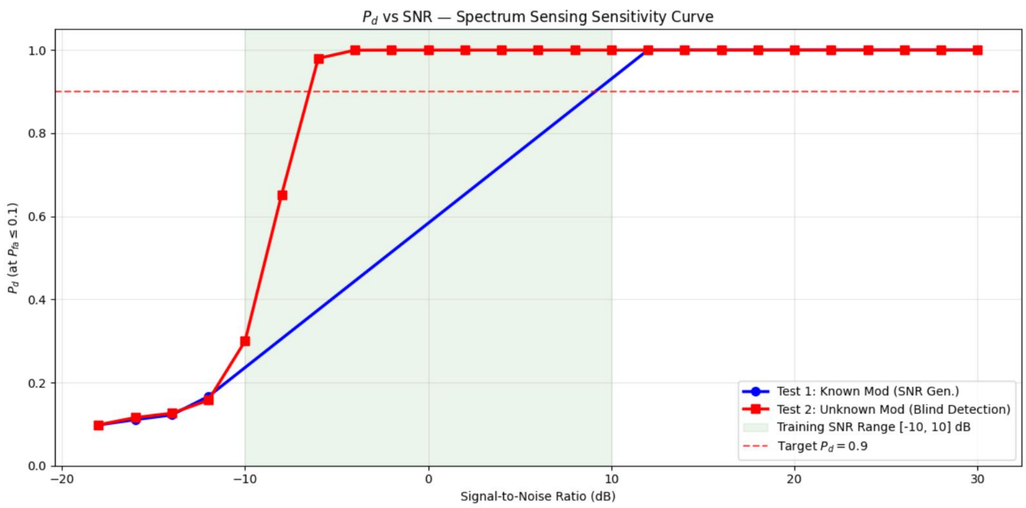
3、初始多模态融合代码

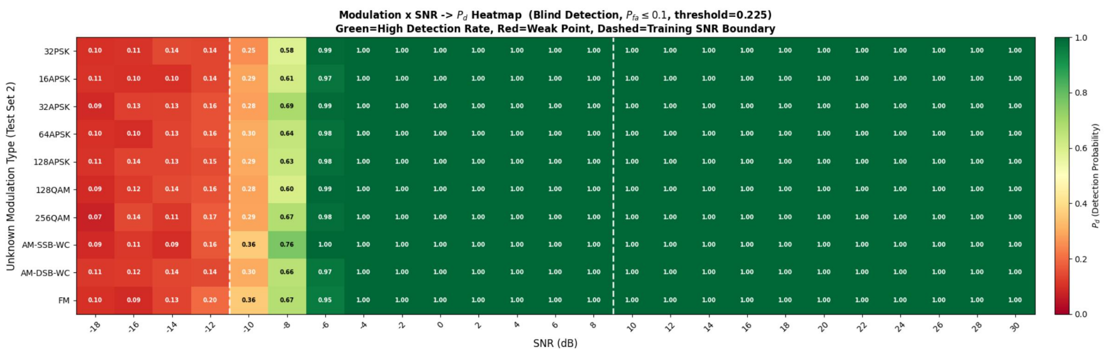
	验证集	Test1	Test2
F1	0.9152	0.8298	0.8658
AUC	0.9626	0.8660	0.9066
Pd	0.8556	0.7205	0.7749
Pfa	0.0141	0.0160	0.0151
ACC	0.9207	0.8522	0.8799
Pre	0.9838	0.9783	0.9808
Rec	0.8556	0.7205	0.7749

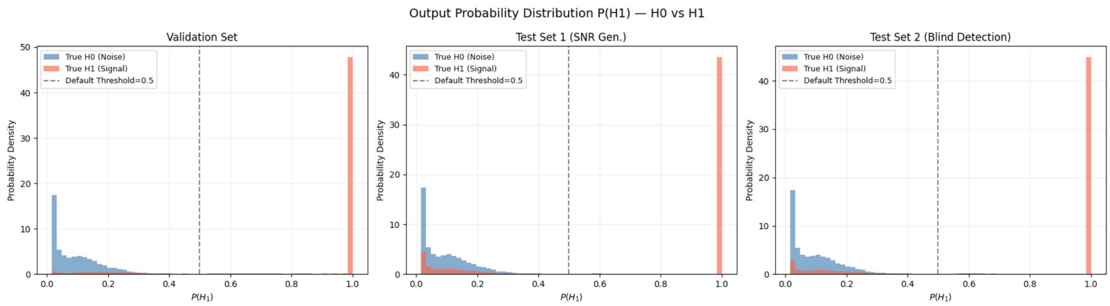

Dual-Branch Multimodal Spectrum Sensing - Training Curves
Loss Curve
Train Loss
Val Loss
CE Loss
Epoch
Accuracy Curve
Train Acc
Val Acc
Accuracy (%)
Epoch


Normalized Confusion Matrices (3-Stage Evaluation)
Validation Set
(Known Mod + Train SNR)
Pd=0.856 Pfa=0.014
0.986
0.014
0.144
0.856
Test Set 1
(SNR Generalization)
Pd=0.720 Pfa=0.016
0.984
0.016
0.280
0.720
Test Set 2
(Modulation Generalization)
Pd=0.775 Pfa=0.015
0.985
0.015
0.225
0.775
True H0
True H1
True Label
Pred H0
Pred H1
Predicted Label


ROC Curves (3-Stage Comparison)
Probability of Detection P_d
Probability of False Alarm P_{fa}
P_{fa} = 0.1 constraint
Validation (AUC=0.9626)
Test 1 (SNR Gen.) (AUC=0.8660)
Test 2 (Blind Detection) (AUC=0.9066)
Random Guess



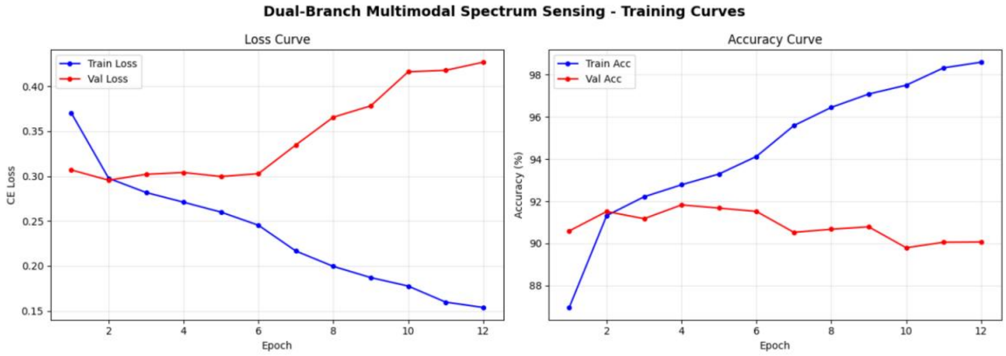


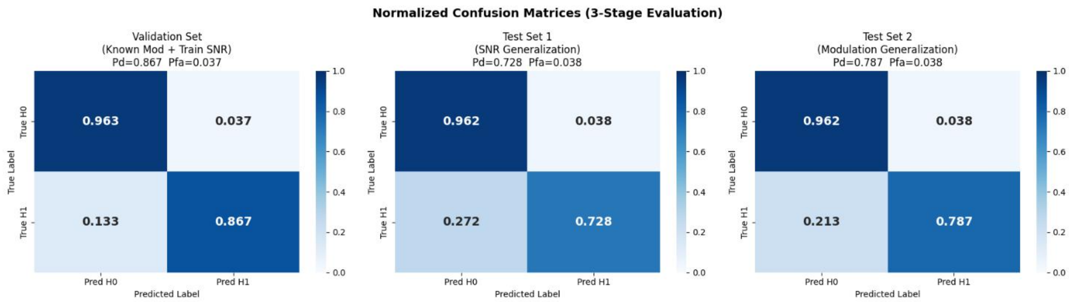


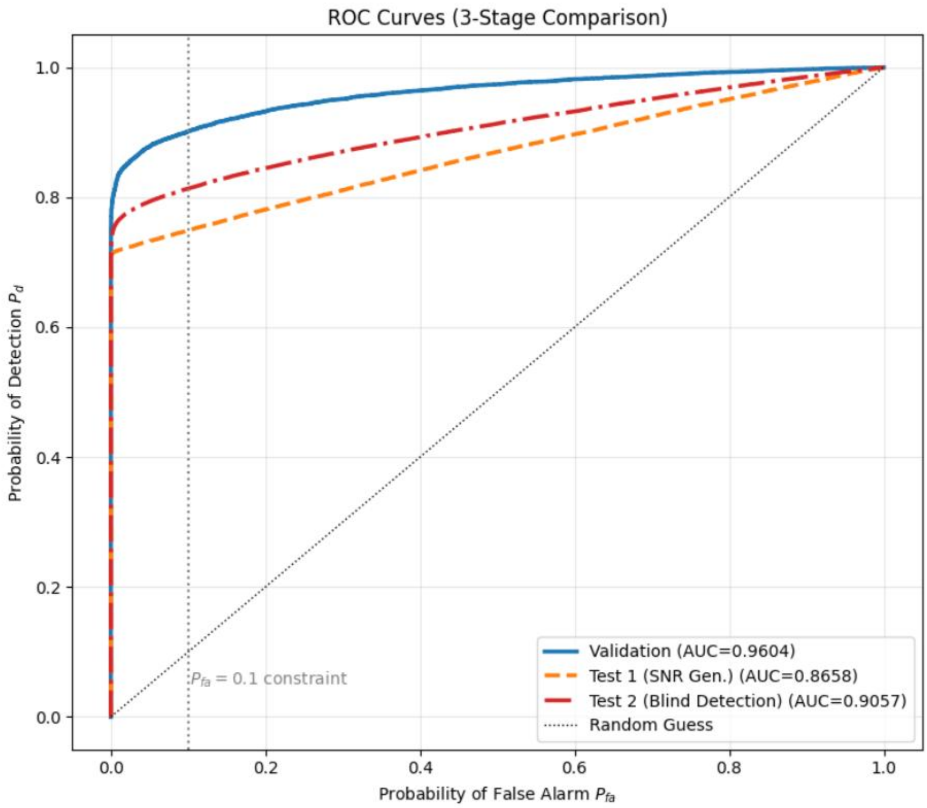

```
4、 阶段 1
├── 2.1 修复 scheduler 重复调用 Bug
├── 3.1 加入早停 + patience=3
├── 3.2 差异化学习率精调
└── 4.1 加入标签平滑 label_smoothing=0.05
    预期提升：Pd +2~3%，过拟合消失
```

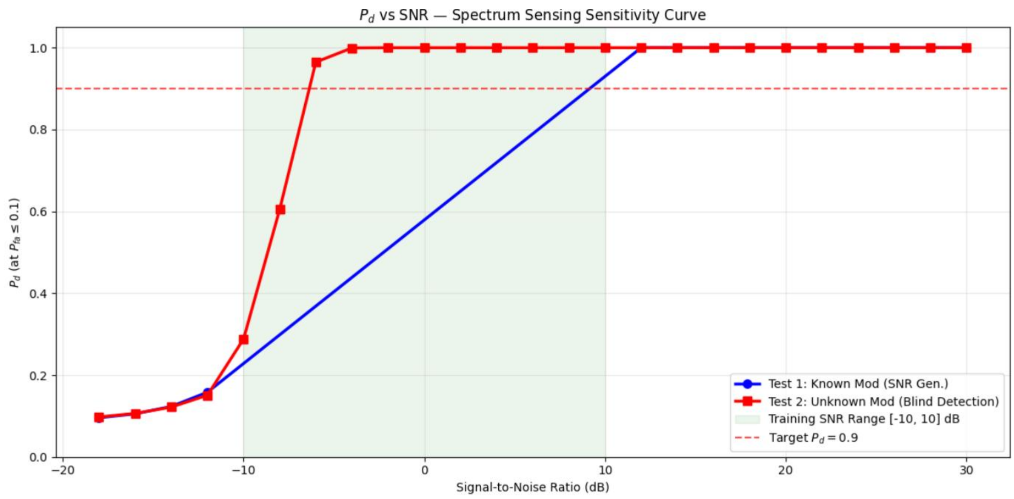
	验证集	Test1	Test2
F1	0.9109	0.8248	0.8626
AUC	0.9604	0.8658	0.9057
Pd	0.8674	0.7238	0.7869
Pfa	0.0372	0.0377	0.0377
ACC	0.9151	0.8453	0.8746

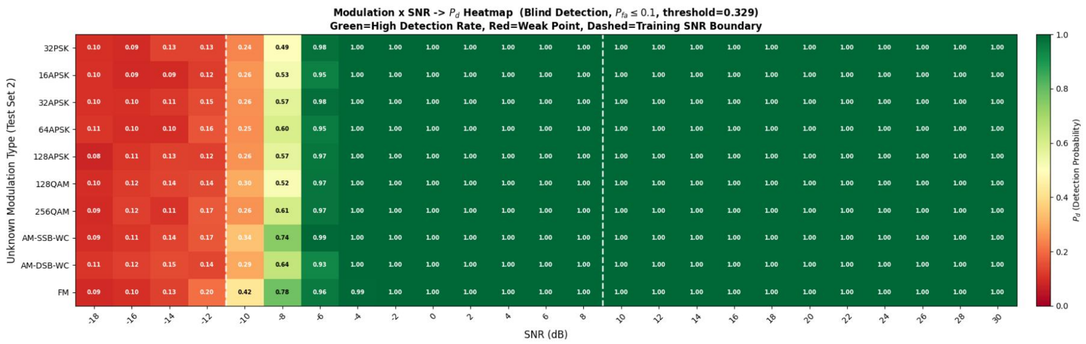
Pre	0.9589	0.9508	0.9543
Rec	0.8674	0.7283	0.7869

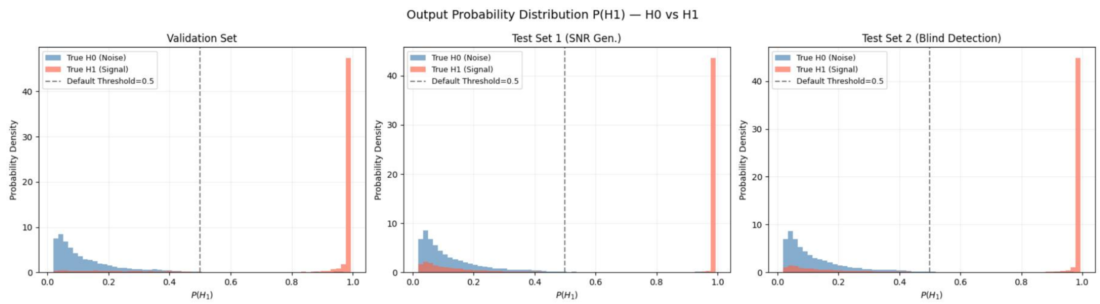












5、 阶段 2（推荐，约 1 小时）

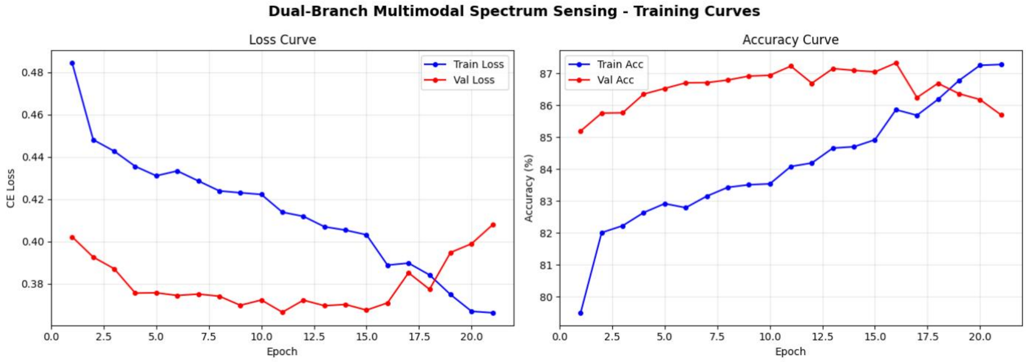
├── 6.1 扩大训练 SNR 到 [-14, +14] dB

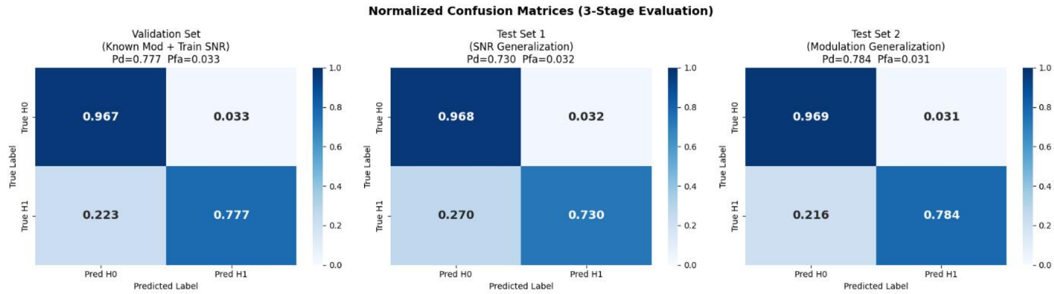
├── 4.3 加入 MixUp 数据增强

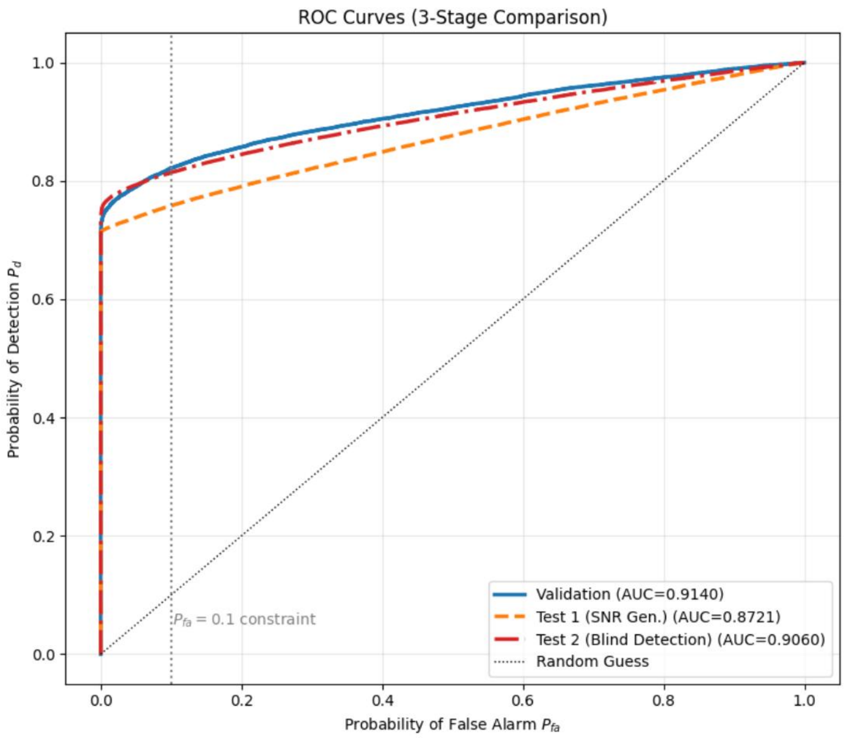
└── 6.3 低 SNR 加噪增强

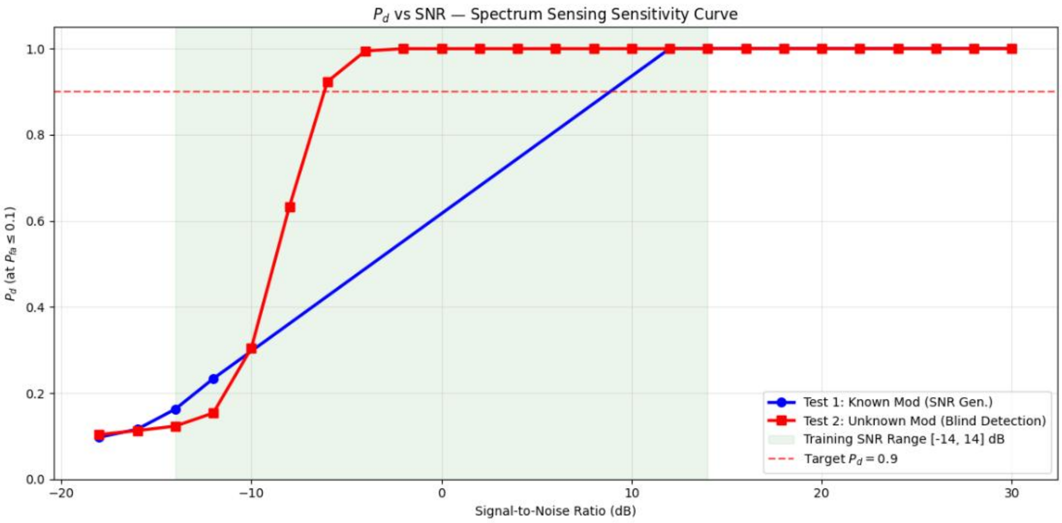
预期提升：测试集 1 Pd +5~8%（低 SNR 专项）

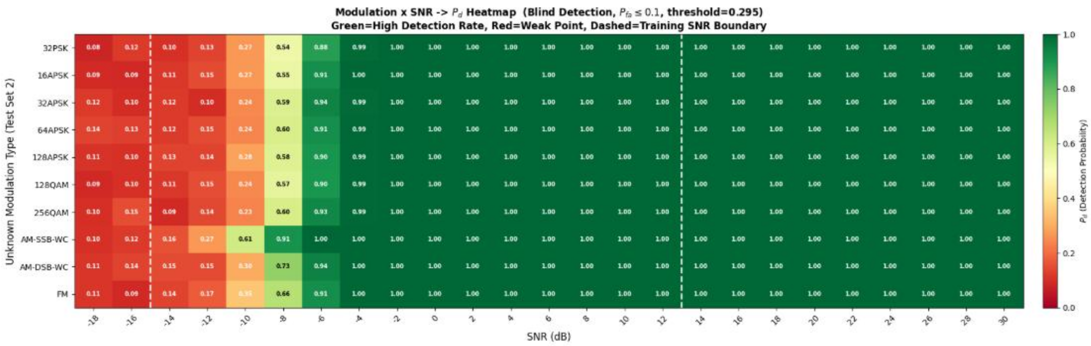
	验证集	Test1	Test2
F1	0.8588	0.8289	0.8638
AUC	0.9140	0.8721	0.9060
Pd	0.7772	0.7303	0.7840
Pfa	0.0326	0.0316	0.0311
ACC	0.8723	0.8493	0.8764
Pre	0.9597	0.9585	0.9618
Rec	0.7772	0.7303	0.7840

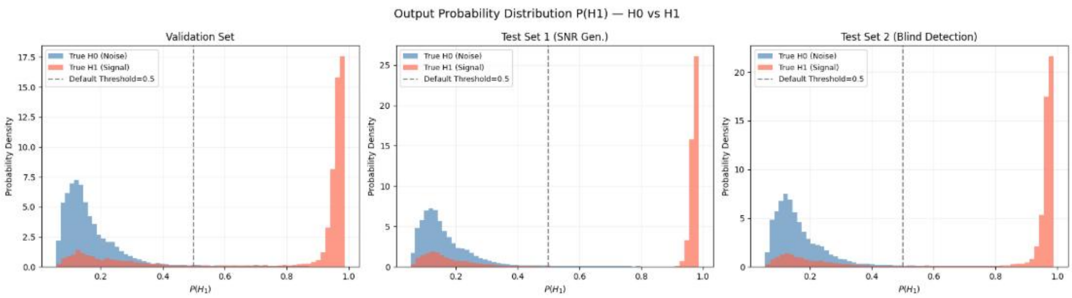

Dual-Branch Multimodal Spectrum Sensing - Training Curves
Loss Curve
Train Loss
Val Loss
CE Loss
Epoch
Accuracy Curve
Train Acc
Val Acc
Accuracy (%)
Epoch


Normalized Confusion Matrices (3-Stage Evaluation)
Validation Set
(Known Mod + Train SNR)
Pd=0.777 Pfa=0.033
0.967
0.033
0.223
0.777
Test Set 1
(SNR Generalization)
Pd=0.730 Pfa=0.032
0.968
0.032
0.270
0.730
Test Set 2
(Modulation Generalization)
Pd=0.784 Pfa=0.031
0.969
0.031
0.216
0.784
True Label
True H0
True H1
Pred H0
Pred H1
Predicted Label


ROC Curves (3-Stage Comparison)
Probability of Detection P_d
Probability of False Alarm P_{fa}
P_{fa} = 0.1 constraint
Validation (AUC=0.9140)
Test 1 (SNR Gen.) (AUC=0.8721)
Test 2 (Blind Detection) (AUC=0.9060)
Random Guess







6、过拟合改善，但是精度不好

（1）删除一个 scheduler.step()；

（2）早停，防止后面记忆训练集：

```
NUM_EPOCHS = 50          # 最大训练轮数
EARLY_STOP = 5           # 连续 5 个 epoch 验证精度无提升则停止；
```

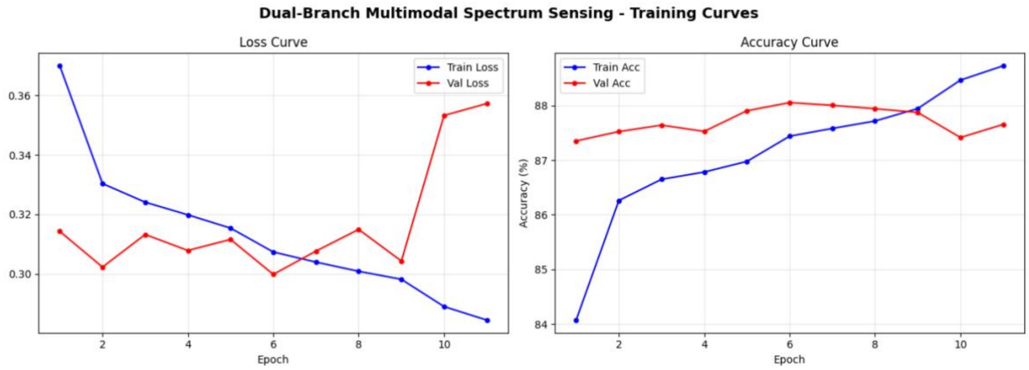
（3）Dropout 增强，p=0.1；

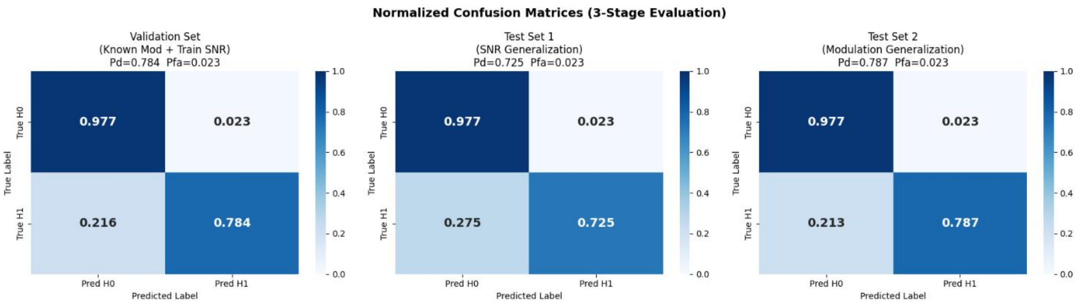
（4）差异化学习率+AdamW：

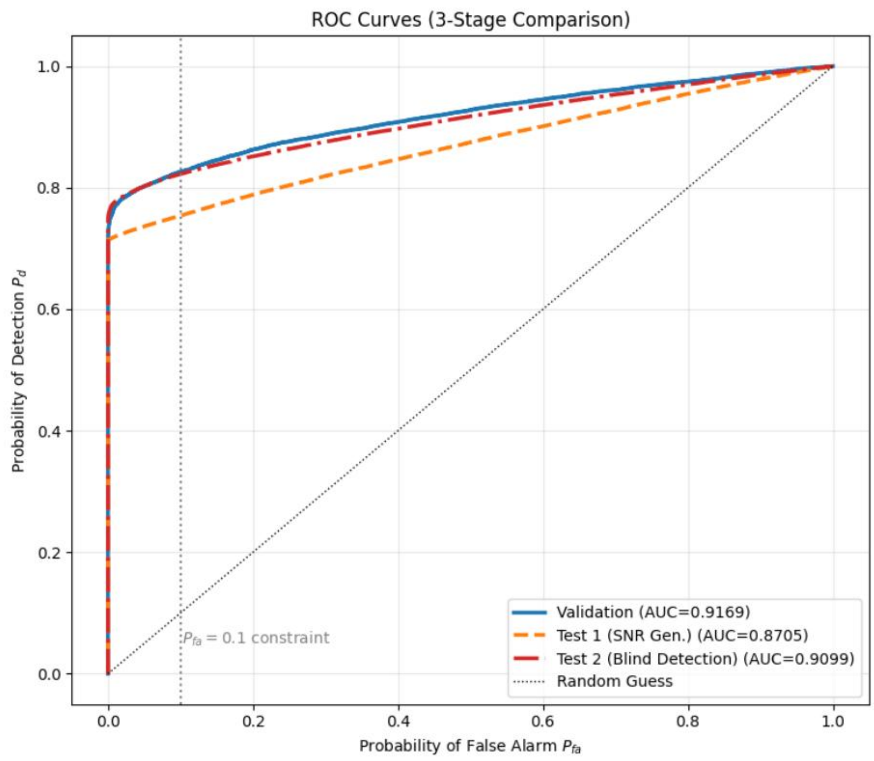
```
optimizer = optim.AdamW([
    {'params': model.time_branch.parameters(),    'lr': 8e-4, 'weight_decay': 3e-4},
    {'params': model.img_branch.parameters(),     'lr': 1e-4, 'weight_decay': 1e-4},
    {'params': model.bca.parameters(),            'lr': 8e-4, 'weight_decay': 3e-4},
    {'params': model.classifier.parameters(),     'lr': 8e-4, 'weight_decay': 3e-4},
```

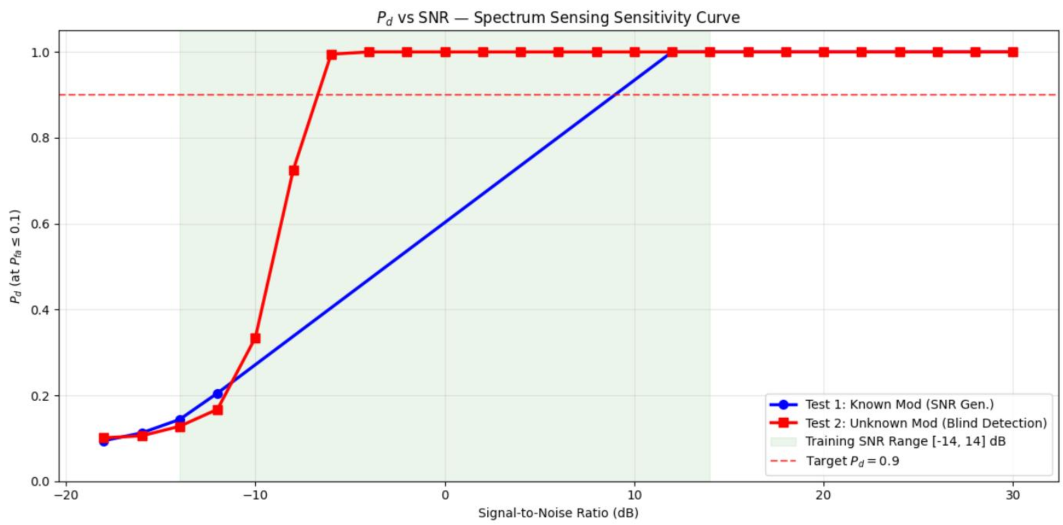
]);
（5）扩大训练 SNR 范围-14—14；
（6）低 SNR 数据增强：加噪增强；
（7）加入通道注意力（CBAM/SE）

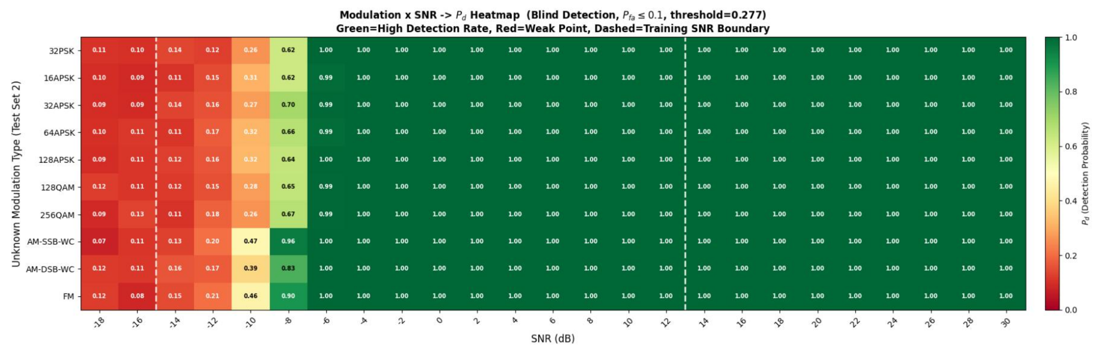
	验证集	Test1	Test2
F1	0.8678	0.8299	0.8697
AUC	0.9169	0.8705	0.9099
Pd	0.7845	0.7252	0.7871
Pfa	0.0234	0.0226	0.0229
ACC	0.8805	0.8513	0.8821
Pre	0.9710	0.9698	0.9717
Rec	0.7845	0.7252	0.7871

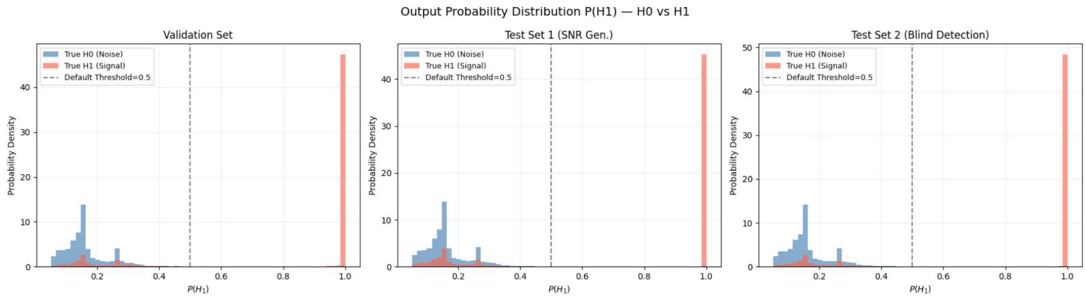





ROC Curves (3-Stage Comparison)
Probability of Detection P_d
Probability of False Alarm P_{fa}
$P_{fa} = 0.1$ constraint
Validation (AUC=0.9169)
Test 1 (SNR Gen.) (AUC=0.8705)
Test 2 (Blind Detection) (AUC=0.9099)
Random Guess


P_d vs SNR — Spectrum Sensing Sensitivity Curve
P_d (at $P_{fa} \leq 0.1$)
Signal-to-Noise Ratio (dB)
Test 1: Known Mod (SNR Gen.)
Test 2: Unknown Mod (Blind Detection)
Training SNR Range [-14, 14] dB
Target $P_d = 0.9$


Modulation x SNR -> P_d Heatmap (Blind Detection, $P_{fa} \leq 0.1$, threshold=0.277)
Green=High Detection Rate, Red=Weak Point, Dashed=Training SNR Boundary
Unknown Modulation Type (Test Set 2)
32PSK
16APSK
32APSK
64APSK
128APSK
128QAM
256QAM
AM-SSB-WC
AM-DSB-WC
FM
SNR (dB)
P_d (Detection Probability)



7、低 SNR Pd 提升，但是验证集和 test2Pd 下降

（1）删除一个 scheduler.step()；

（2）早停，防止后面记忆训练集：

NUM_EPOCHS = 50   # 最大训练轮数

EARLY_STOP = 5   # 连续 5 个 epoch 验证精度无提升则停止；

（3）扩大训练 SNR 范围-14—14；

（4）Test2 只测试未知调制泛化   TEST2_H1_SNRS = TRAIN_H1_SNRS；

（5）只对低 SNR 的 H1 以 15%的概率轻微降低 1-4db

```
if self.training and y == 1 and snr <= -8 and np.random.rand() < 0.15:
    extra_snr_drop = np.random.uniform(1, 4)
```

（6）加入低 SNR H1 过采样

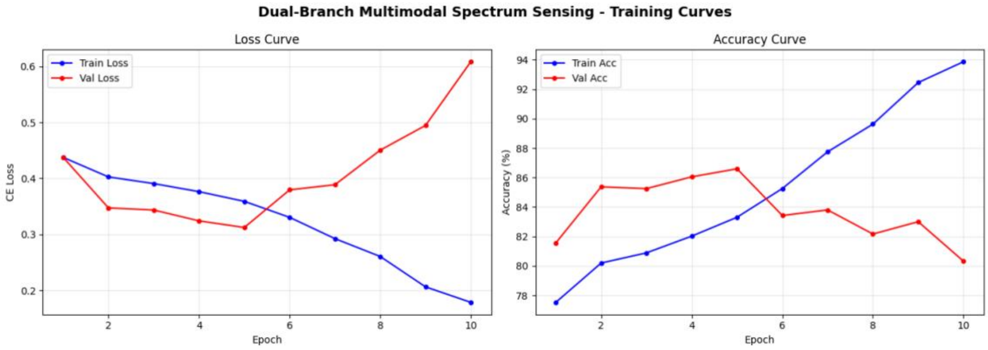
（7）通道注意力

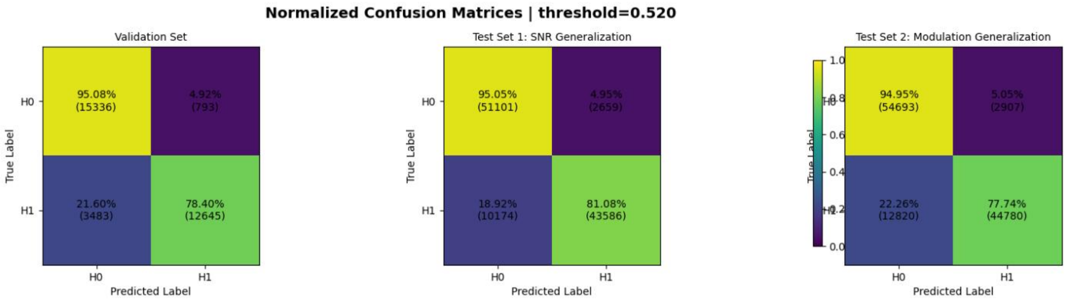
（8）降低 weight_decay 和 BCA 前的 Dropout

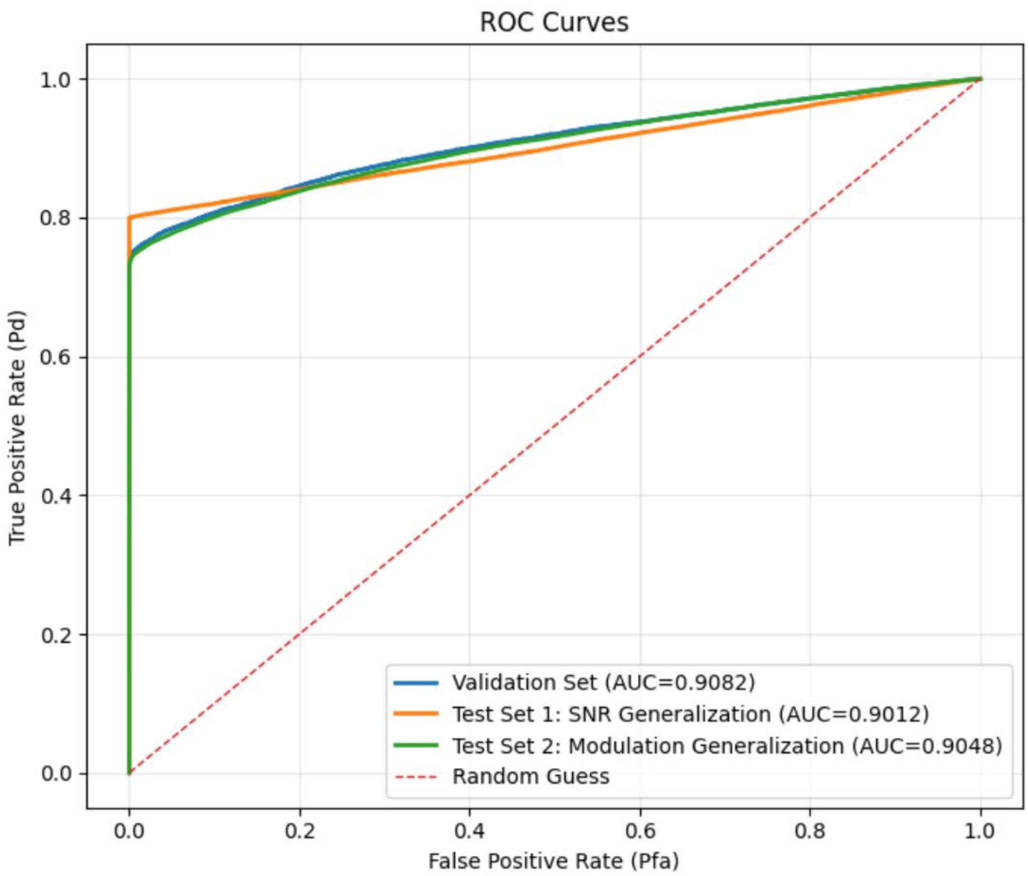
（9）改成阈值优化评估

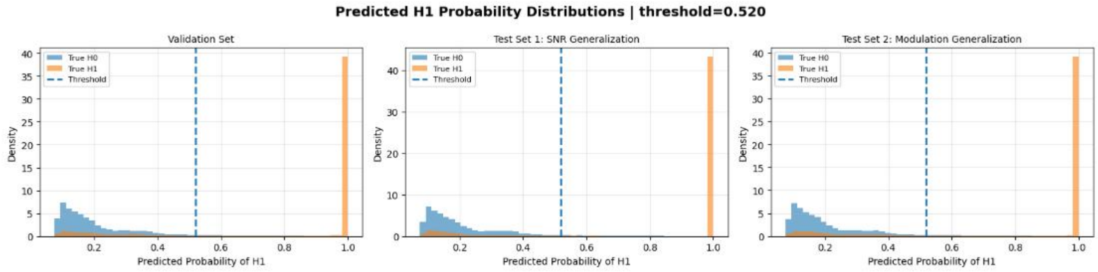
```
[1] Accuracy 最优阈值: threshold = 0.743
    Val Acc = 0.8732
    Val Pd  = 0.7563
    Val Pfa = 0.0099
    Val AUC = 0.9082
[2] Pfa≤5% 条件下 Pd 最优阈值: threshold = 0.520
    Val Acc = 0.8674
    Val Pd  = 0.7840
    Val Pfa = 0.0492
    Val AUC = 0.9082
☑最终采用阈值 FINAL_THRESHOLD = 0.520
```

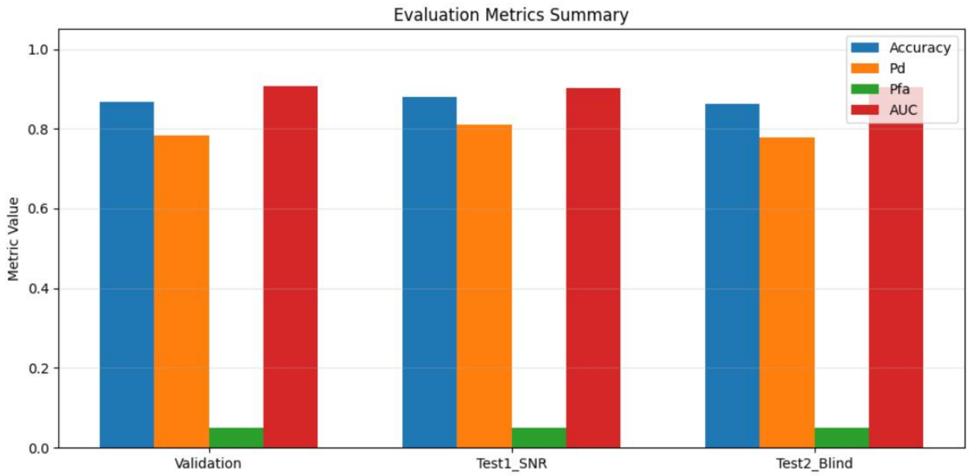
	验证集	Test1	Test2
F1	0.8554	0.8717	0.8506
AUC	0.9082	0.9012	0.9048
Pd	0.7840	0.8108	0.7774
Pfa	0.0492	0.0495	0.0505
ACC	0.8674	0.8806	0.8635
Pre	0.9410	0.9425	0.9390
Rec	0.7840	0.8108	0.7774


Dual-Branch Multimodal Spectrum Sensing - Training Curves
Loss Curve
Train Loss
Val Loss
CE Loss
Epoch
Accuracy Curve
Train Acc
Val Acc
Accuracy (%)
Epoch


Normalized Confusion Matrices | threshold=0.520
Validation Set
95.08% (15336)
4.92% (793)
21.60% (3483)
78.40% (12645)
Test Set 1: SNR Generalization
95.05% (51101)
4.95% (2659)
18.92% (10174)
81.08% (43586)
Test Set 2: Modulation Generalization
94.95% (54693)
5.05% (2907)
22.26% (12820)
77.74% (44780)
True Label
Predicted Label
H0
H1


ROC Curves
True Positive Rate (Pd)
False Positive Rate (Pfa)
Validation Set (AUC=0.9082)
Test Set 1: SNR Generalization (AUC=0.9012)
Test Set 2: Modulation Generalization (AUC=0.9048)
Random Guess


Predicted H1 Probability Distributions | threshold=0.520
Validation Set
Test Set 1: SNR Generalization
Test Set 2: Modulation Generalization
True H0
True H1
Threshold
Density
Predicted Probability of H1


Evaluation Metrics Summary
Accuracy
Pd
Pfa
AUC
Metric Value
Validation
Test1_SNR
Test2_Blind